Activity 2.4 – Finite State Machine Design

1.Design Approach:

Designed a finite state machine using Verilog HDL with three states: IDLE, WORK and DONE. Implemented sequential logic to control state transitions based on start and done input signals.

2.Verification Steps:

- Created a separate testbench.
- Applied reset to initialize the FSM.
- Activated the start signal.
- Verified transition from IDLE to WORK.
- Activated the done signal.
- Checked the return to IDLE.
- Simulated using ModelSim.

3.Results:

The testbench checks state transitions and busy signal generation. PASS/FAIL messages are displayed.

4.Issues Faced:

State transitions must be properly defined. Reset conditions must initialize the FSM correctly. Output signals must correspond to the current state.

5.Conclusion:

Successfully designed and verified a finite state machine using Verilog HDL.